HT_MK1 Harness Tester

GUI Functional Description

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1 Introduction

1.1 Purpose

This document describes what the HT_MK1 Console (the GUI) does, screen by screen and button by button: what each area shows, what each control does when it is pressed, when it is enabled or disabled, and what message or state follows. It is the reference to use when you want to know exactly what a particular button, pill, table or dialog is for.

1.2 Audience

Operators and test engineers who use the HT_MK1 Console to test harnesses, and engineers who evaluate or accept the software. For step-by-step operating instructions, use the HT_MK1 User Manual instead; this document explains the software itself.

1.3 Conventions

- Button, pill, tab and column names are written in **bold**. On screen the GUI draws most of them in capital letters; this document writes them in normal case.
- Text the GUI shows as a message is written in `monospace`.
- Connector names: **J-MTX** is the Matrix connector (continuity, resistance, auto scan); **J-HV** is the HV connector (insulation). When the HV stack is N cards the GUI writes the HV connector as **J-HV1** (one card) or **J-HV1-N**.
- Mode 1 means continuity and resistance on J-MTX. Mode 2 means the insulation test on J-HV. Auto scan is a separate survey that needs no netlist.

1.4 Related Documents

- HT_MK1 User Manual: step-by-step operation and safety.
- HT_MK1 Functional Test Procedure: the formal acceptance test of a built unit.
- HT_MK1 Flashing Procedure: programming the firmware.
- HT_MK1 Code Walkthrough: software structure and the serial protocol.
- HT_MK1 System Overview: what the instrument measures and how.

2 Application Overview

2.1 What the GUI is

The HT_MK1 Console is a Windows desktop application. It contains no test logic of its own: it sends commands to the instrument over a serial link, receives the results as they are measured, and presents them. Everything that decides pass or fail, and everything that switches high voltage, is done by the instrument firmware.

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2.2 Program files

File	Use
HT_MK1_GUI.exe	The operator GUI for the real instrument.
HT_MK1_GUI_Demo.exe	The same GUI with a built-in simulator and no instrument. It is for practice and training. Nothing it shows is a measurement.

Table 1: Program files

Both programs are single files that unpack to a temporary folder each time they start, so the first window takes a second or two. Nothing is installed. They run on 64-bit Windows 10 and 11.

2.3 Window

The window opens maximized. Its title is HT_MK1 Console. The GUI follows the Windows light or dark setting at start-up; the sun/moon button in the status bar switches between them and the choice is remembered.

2.4 Build stamps

The footer shows when the GUI was built (`GUI build YYYY-MM-DD HH:MM`) and, once the instrument has answered, when its firmware was built. The status bar shows the firmware version (`fw 1.0.0`).

3 Window Layout

Every screen has the same four regions, from top to bottom: the status bar, the middle area (navigation rail on the left and the screen itself on the right), the log bar, and the footer.

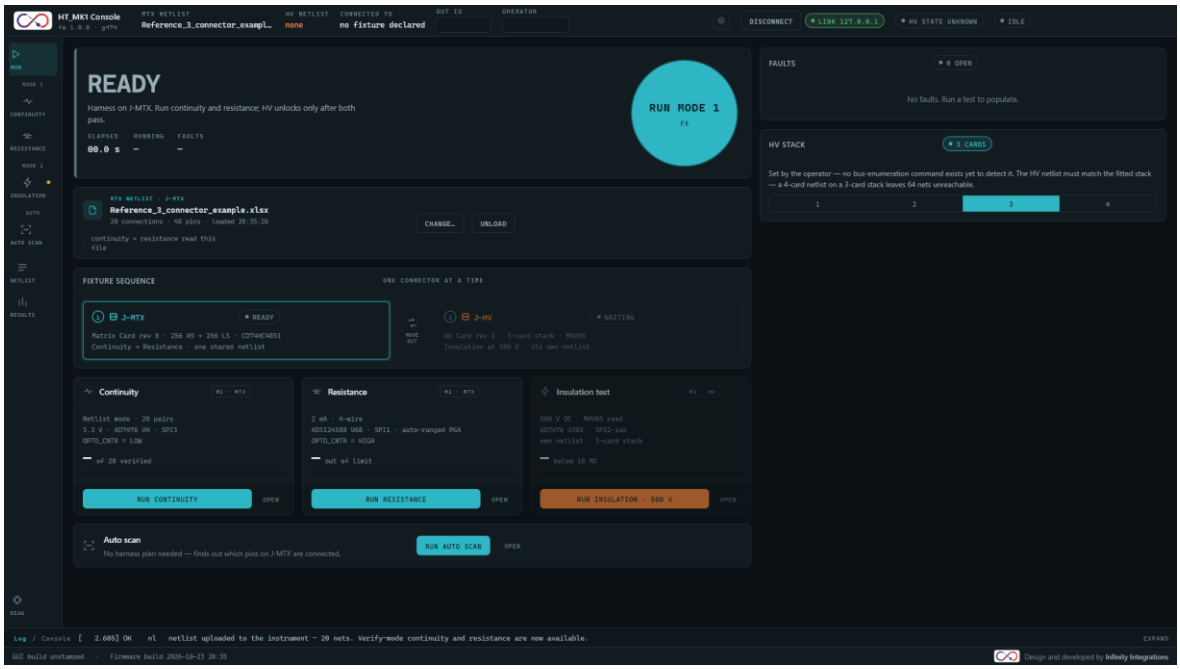


Figure 1: The Run screen with a netlist loaded: status bar (top), navigation rail (left), screen (centre and right), log bar and footer (bottom).

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Region	Shows
Status bar	Connection, loaded netlists, which connector the harness is on, DUT ID and Operator fields, theme button, serial port controls and the HV and state pills (Section 4).
Navigation rail	The seven screens, grouped by mode (Section 5).
Screen	The selected screen. It scrolls when it is taller than the window.
Log bar	The most recent log line; expands to a scrolling list (Section 14).
Footer	Build stamps and the credit line (Section 14).

Table 2: Window regions

When the instrument reports high voltage on the HV rail, the status bar is replaced by the high-voltage banner (Section 4.4) so that the hazard cannot be missed. In a narrow window (about 860 pixels or less) the navigation rail becomes a horizontal strip under the header, and below about 1100 pixels the status bar scrolls sideways and the build stamps are hidden.

4 Status Bar

4.1 Fields, left to right

Item	What it shows or does
Logo and name	The company logo, then (in windows wider than 1200 pixels) `HT_MK1 Console` and the firmware version, for example `fw 1.0.0 · g474`. The version reads `fw -` until the instrument has answered.
MTX netlist	File name of the netlist loaded for continuity and resistance, or `none`.
HV netlist	File name of the netlist loaded for insulation, or `none`.
Connected to	Where the instrument believes the harness is: `J-MTX · mode 1`, `J-HV · mode 2`, `no fixture declared`, `none · moving` (during handover), `J-HV1-N · mode 2` (after the harness move is confirmed) or `link lost`.
DUT ID	Free text. Identifies the unit under test. It is copied into every report made afterwards. It is not saved between sessions.
Operator	Free text. Name of the person running the test. Copied into reports, not saved between sessions.
Theme button	Switches between light and dark.
Port controls	Serial port list, refresh button, Connect/Disconnect button and link pill (Section 4.2).
HV pill	High-voltage state, taken only from what the instrument reports (Section 4.3).
State pill	What the instrument or the test sequence is doing (Section 4.3).
Clear Fault	Appears only while the instrument is in the fault state (Section 16.5).

Table 3: Status bar items

Note: The DUT ID and Operator fields are drawn in the status bar only when the window is wider than 1600 logical pixels. On a smaller or heavily scaled display the same two fields are shown in the Test details panel at the top of the right column of the Run screen (Section 6.9).

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4.2 Serial port controls

Control	Behaviour
Port list	Shows the selected port, `Select port` when none is chosen, or `No ports` when none were found. It is enabled only while not connected. Each entry reads the port name and its description, for example the ST-LINK Virtual COM Port. Choosing an entry only selects it; it does not connect.
Refresh	Looks for serial ports again. The log reports the ports found, or that none were found. If the selected port has disappeared, the first available port is selected.
Connect	Opens the selected port. It is disabled until a port is selected. While opening, the button reads `Connecting` and is disabled. On success the GUI reads the instrument state, identity, calibration, limits and any netlist already held by the instrument.
Disconnect	Replaces Connect while connected. It closes the link; the GUI then shows the state as unknown.
Link pill	`Link <port>` (green) when connected, `Opening <port>` (blue) while connecting, `No link` (amber) otherwise.

Table 4: Serial port controls

When the GUI is started with the built-in simulator (the Demo program) the port list and refresh button are not shown, because the link is internal.

4.3 HV pill and state pill

HV pill text	Colour	Meaning
HV Safe	Green	The instrument has reported that the system is safe, and nothing since has contradicted it. This is the only indication that it is safe to handle the harness.
HV armed	Amber	The instrument is armed for an insulation run. High voltage is not yet applied.
HV LIVE n V	Red	The instrument reports the HV rail at or above 50 V.
HV state unknown	Grey	Just connected, a fault is latched, a test has run and no safe report has arrived yet, or any other case where safety has not been confirmed.
Link lost	Amber	The connection to the instrument is down. Treat the system as not safe.

Table 5: HV pill values

State pill text	Meaning
Ready	The GUI is reset and waiting for a test (initial state, and after a reset).
Idle	The instrument reports idle.
Cont, Res, Insul (blue)	A continuity, resistance or insulation run is in progress.
Fail (red)	Mode 1 finished with a failure.
Handover (amber)	Mode 1 passed; the harness must now be moved to J-HV.
Armed (amber)	The harness move was confirmed; insulation is unlocked.
Hv_armed	The instrument itself reports the armed state.
Fault (red)	The instrument has latched a fault. Press Clear Fault after resolving the cause.
State unknown (amber)	The link is down.

Table 6: State pill values

4.4 High-voltage banner

While the instrument reports the HV rail at or above 50 V, the status bar is replaced by a striped hazard banner. It shows a pulsing lightning bolt, the headline `HIGH VOLTAGE LIVE - J-HV ENERGIZED`, the line `Insulation test running. Relays are cold-switched; the fixture stays live until discharge completes.`, the live **Rail** voltage, the live **Leakage** voltage and an **Abort & discharge** button.

- **Abort & discharge** sends the abort command to the instrument, writes `ABORT sent` to the log and then resets the GUI to its ready state. The instrument is responsible for removing the voltage and then reports when it is safe.

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- The banner disappears when the rail falls below 50 V, after an abort, and when the link is lost. After link loss the status bar returns and shows `Link lost` and `State unknown`; do not assume the harness is safe.
- The other status bar controls are not reachable while the banner is shown.

5 Navigation Rail

The rail on the left switches between screens. Selecting an item always works; no screen is locked.

Group	Item	Screen
(none)	Run	The home screen: verdict, big run button, fixture sequence, test cards, faults.
Mode 1	Continuity	Continuity on J-MTX.
Mode 1	Resistance	Resistance on J-MTX.
Mode 2	Insulation	Insulation test on J-HV.
Auto	Auto scan	Finds which pins are connected, with no netlist.
(none)	Netlist	The two netlists and what they contain.
(none)	Results	Pass rate, run history and report export.

Table 7: Navigation rail items

A small dot on the icon of Continuity, Resistance and Insulation summarizes that test:

- A **green** dot: the last result of that test was a pass.
- A **red** dot: the last result was a fail.
- An **amber** dot (Insulation only): the harness has not yet been declared moved to J-HV and Insulation has no result, that is, insulation is still locked.
- No dot: the test has not been run.

All dots are cleared whenever the GUI is reset.

6 Run Screen

The Run screen is the home screen. It has a left column with the verdict and the test controls, and a right column with the fault list and the HV stack setting. In a narrow window the right column moves below the left one.

6.1 Verdict panel

The large panel at the top left shows the overall state. A coloured rule on its left edge is grey when idle, blue while running, amber at the handover, HV colour when armed, green for a pass and red for a fail.

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Headline	Sub-text
READY	`Harness on J-MTX. Run continuity and resistance; HV unlocks only after both pass.`
RUNNING	A line naming the test, for example `J-MTX - testing the pairs the netlist expects.` or `J-HV - 500 V across each net in turn.`
PARTIAL	One of the two Mode 1 tests has not run: `Continuity has not run yet. Both must pass before HV unlocks.` (or the same for Resistance).
FAIL	`Mode 1 failed on J-MTX. The harness stays where it is - HV is not attempted.`
MOVE DUT	`Continuity and resistance passed on J-MTX. Move the harness to the HV fixture.`
ARMED	`Harness on J-HV1-N. Press Run Insulation - you will be asked to confirm before 500 V is applied.`

Table 8: Verdict headlines

Under the headline are three values: **Elapsed** (`00.0 s` when idle; while a test runs it shows progress as `done / total`, for example `37 / 128`), **Running** (the stage: `Continuity · J-MTX`, `Resistance · J-MTX`, `Insulation · J-HV`, `handover` or `HV ready`) and **Faults** (`-` when idle, otherwise the number of open fault entries).

6.2 Big round button

The large round button on the right of the verdict panel is the main control. Its label and effect depend on the state. The keyboard shortcut F5 does the same as pressing it, but only when no test is running.

Label (hint)	When it appears	What pressing it does
Stop (Esc)	A test is running.	Sends the abort command to the instrument, then resets the GUI (see Section 16.3).
Move DUT (see below)	Continuity and resistance have both passed and the harness has not been declared moved.	Disabled. Use the Harness moved button in the handover panel.
Run Insul (500 V)	The instrument reports the harness is on J-HV and insulation has no result.	Starts the HV start sequence (Section 13.3).
Run Mode 1 (F5)	Neither Mode 1 test has a result.	Starts continuity. If netlist mode is selected and no MTX netlist is loaded, opens the Select the MTX netlist dialog instead.
Reset (Esc)	Any other case.	If continuity passed and resistance has not run, starts resistance; otherwise resets the GUI. With no link to the instrument it does nothing except log `not connected: choose the port and press Connect first`.

Table 9: Big round button

Note: Run Mode 1 starts continuity only. After continuity passes, run resistance with Run resistance (or the big button, which then reads Reset but starts resistance). The two tests are not chained automatically.

6.3 Netlist strip

Below the verdict panel, a strip shows the MTX netlist. Every screen that needs a netlist shows the same strip. See Section 11.1 for its buttons.

6.4 Handover panel

When continuity and resistance have both passed, an amber striped panel appears: `Move the harness to the HV fixture` with the text `Continuity and resistance passed on J-MTX. Unplug from J-MTX, then plug into J-HV1-N.` and the button **Harness moved - unlock HV**.

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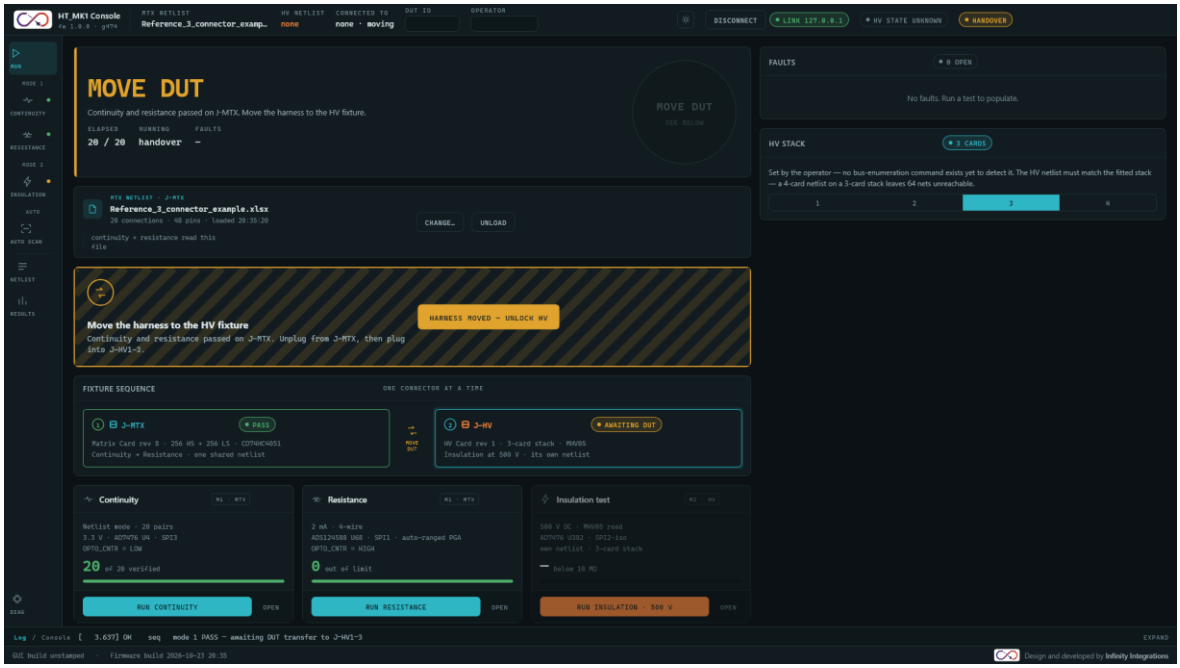


Figure 2: The Run screen after Mode 1 passed: the verdict reads MOVE DUT and the handover panel asks for the harness to be moved.

Pressing **Harness moved - unlock HV** tells the instrument the harness is on J-HV (a fixture command), then the GUI moves to the armed state: verdict ARMED, state pill Armed, Connected to `J-HV1-N · mode 2`, and the Insulation card becomes active. The log records `DUT transferred to J-HV1-N - HV armed, interlock closed`. If the instrument refuses the fixture change, the refusal is written to the log and the GUI stays at the handover.

6.5 Fixture sequence panel

This panel shows the two connectors in order, with a MOVE DUT arrow between them (amber when the handover is awaited).

Box	Contents	Pill values
1 J-MTX	256 HS + 256 LS. Continuity, then resistance, sharing one netlist.	Ready, Pass (green), Fail (red), Partial (blue), Released (grey, after the handover).
2 J-HV	The N-card stack. Insulation at 500 V with its own netlist.	Waiting (grey, dimmed), Awaiting DUT (amber), Armed (amber).

Table 10: Fixture sequence boxes

6.6 Test cards

Three cards, one per test, sit side by side (they wrap when the window is narrow). Each card has a title and a stage badge, a condition text, a large count with its unit, a progress bar, a Run button and an **Open** link that jumps to that test's own screen.

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Card	Badge	Condition text	Large number	Run button
Continuity	M1 · MTX	Netlist mode: `Netlist mode · N pairs.` Cross mode: `Cross continuity · 65,536 reads.`	Pairs passed so far, with `of N verified` or `nets discovered.`	Run continuity
Resistance	M1 · MTX	`2 mA · 4-wire`	Pairs out of limit so far.	Run resistance
Insulation test	M2 · HV	`500 V DC` and `own netlist · N-card stack`	Nets below the limit so far, with the unit `below 10 MΩ.`	Run insulation · 500 V

Table 11: Test cards

The card colours follow the result: green for a pass, red for a fail, blue while running. The Insulation card is dimmed until the handover has been confirmed. **Run continuity** and **Run resistance** are enabled only when the rules in Section 6.10 allow it. **Run insulation · 500 V** is enabled whenever no test is running; when the harness is not yet on J-HV it opens the prerequisites dialog (Section 13.4) instead of blocking.

6.7 Auto scan strip

A strip with the text `No harness plan needed - finds out which pins on J-MTX are connected.` and the button **Run auto scan**, which opens the Auto Scan screen and starts the scan. It is disabled while a test runs or when the harness is on J-HV, and shows the reason in place of its text. **Open** jumps to the Auto Scan screen.

6.8 Faults panel

The right column starts with the Faults panel. Its pill shows `0 open` (grey) or `N open` (red). When empty it reads `No faults. Run a test to populate.` Otherwise there is one row per fault type, listing only the first failing net of each type. Pressing a row jumps to the screen that shows the fault.

Code	Title	Detail	Jumps to
F06	Open circuit	Continuity, net name, source and destination pins	Continuity, with the wiring filter set to Faults only
F08	Resistance high	Resistance, net name and limit; the value is the measured ohms	Resistance
F04	Insulation below limit	HV, card, relay, net name, 500 V; the value is the insulation in MΩ	Insulation (when the harness is on HV), with that relay marked red

Table 12: Fault codes shown in the Faults panel

6.9 Test details panel

On a window that is not wide enough for the status bar to carry the DUT ID and Operator fields (1600 logical pixels or less), the first panel of the right column is **Test details**, with the same two text boxes. Both are copied into every report made afterwards and are not saved between sessions.

6.10 HV stack panel

The HV stack panel states how many HV cards are fitted (1 to 4, default 3). The instrument cannot detect the stack size, so the operator sets it by pressing **1**, **2**, **3** or **4**. The choice is not sent to the instrument. It sets the HV connector name (J-HV1 to J-HV1-N), which card rows count as fitted on the Insulation screen, which nets are reachable in the HV netlist table and the HV netlist check in the pre-start dialog. The text on the panel

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reminds you that the HV netlist must match the fitted stack: a 4-card netlist on a 3-card stack leaves 64 nets unreachable.

Important: Set the HV stack to the number of HV cards actually fitted in the instrument before running insulation.

6.11 When the Run buttons are enabled

Control	Blocked when	Message shown beside the button
Any run control	A test is already running.	`A test is already running`
Any run control	There is no link to the instrument.	`Not connected: choose the port and press Connect first`
Run continuity	The harness is on J-HV.	`Harness is on J-HV - reset to return it to J-MTX`
Run continuity	Netlist mode and no MTX netlist is loaded.	`Netlist mode needs the MTX netlist - or switch to cross continuity`
Run resistance	The harness is on J-HV, or no MTX netlist is loaded.	`Resistance needs the MTX netlist for the pair list and limits`
Run insulation (on J-HV)	The harness has not been confirmed on J-HV.	`Confirm the harness has been moved to J-HV first` or `Continuity and resistance must pass before HV unlocks`
Run auto scan	A test is running, or the harness is on J-HV.	The same reasons as above

Table 13: Run control gating

Note: Without a link to the instrument no Run control is enabled. Each shows the reason `Not connected: choose the port and press Connect first`. Connect first (Section 4.2).

7 Continuity Screen

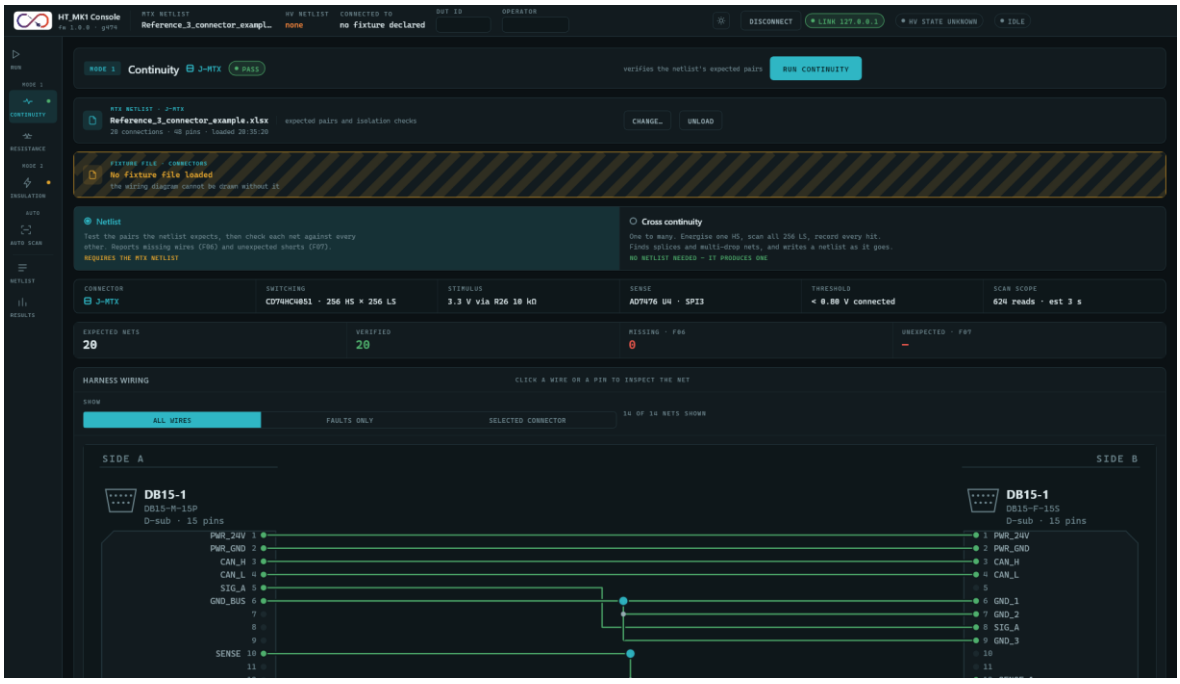


Figure 3: The Continuity screen after a passing netlist-mode run.

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7.1 View bar

The first panel shows the stage badge **Mode 1**, the heading **Continuity**, the connector chip **J-MTX**, a status pill (`Not run`, `Running`, `Pass`, `Fail`), the reason text of Section 6.10 and the **Run continuity** button.

7.2 Netlist and fixture strips

The MTX netlist strip (Section 11.1) sits under the view bar. Under it, a **Fixture file** strip is reserved for a future connector-file feature. It currently always reads `No fixture file loaded`; the connector layout used for the wiring diagram comes from the netlist (Section 13.5), so this strip does not stop the diagram from being drawn.

7.3 Mode selector

Two options choose what the next run does. Selecting one takes effect immediately, even while a test is running, but it changes only the next run and which panels are shown.

Option	Description	Requirement tag
Netlist	Test the pairs the netlist expects, then check each net against every other. Reports missing wires (F06) and unexpected shorts (F07).	REQUIRES THE MTX NETLIST (amber)
Cross continuity	One to many. Energise one HS pin, scan all 256 LS pins and record every hit. Finds splices and multi-drop nets and writes a netlist as it goes.	NO NETLIST NEEDED, IT PRODUCES ONE (green)

Table 14: Continuity modes

7.4 Facts band

A row of fixed facts: **Connector** `J-MTX`; **Threshold** `< 0.80 V connected`; **Scan scope** `624 reads, est 3 s` (netlist mode) or `65,536 reads, est 18 s` (cross mode). The scan scope values are fixed text, not a calculation for the loaded harness.

7.5 Result tiles

Mode	Tile 1	Tile 2	Tile 3	Tile 4
Netlist	Expected nets	Verified (green)	Missing · F06 (red)	Unexpected · F07 (red, shows a dash)
Cross	Nets discovered	Nodes mapped	Multi-drop	HS columns swept

Table 15: Continuity result tiles

7.6 Harness wiring diagram

In netlist mode, the Harness wiring panel draws the harness as two columns: **Side A** (the HI pins, from the Conn ID columns of the netlist) on the left and **Side B** (the LO pins, from the Conn ID B columns) on the right. Before a netlist is loaded the panel says `No harness to draw yet`. It is not shown in cross mode.

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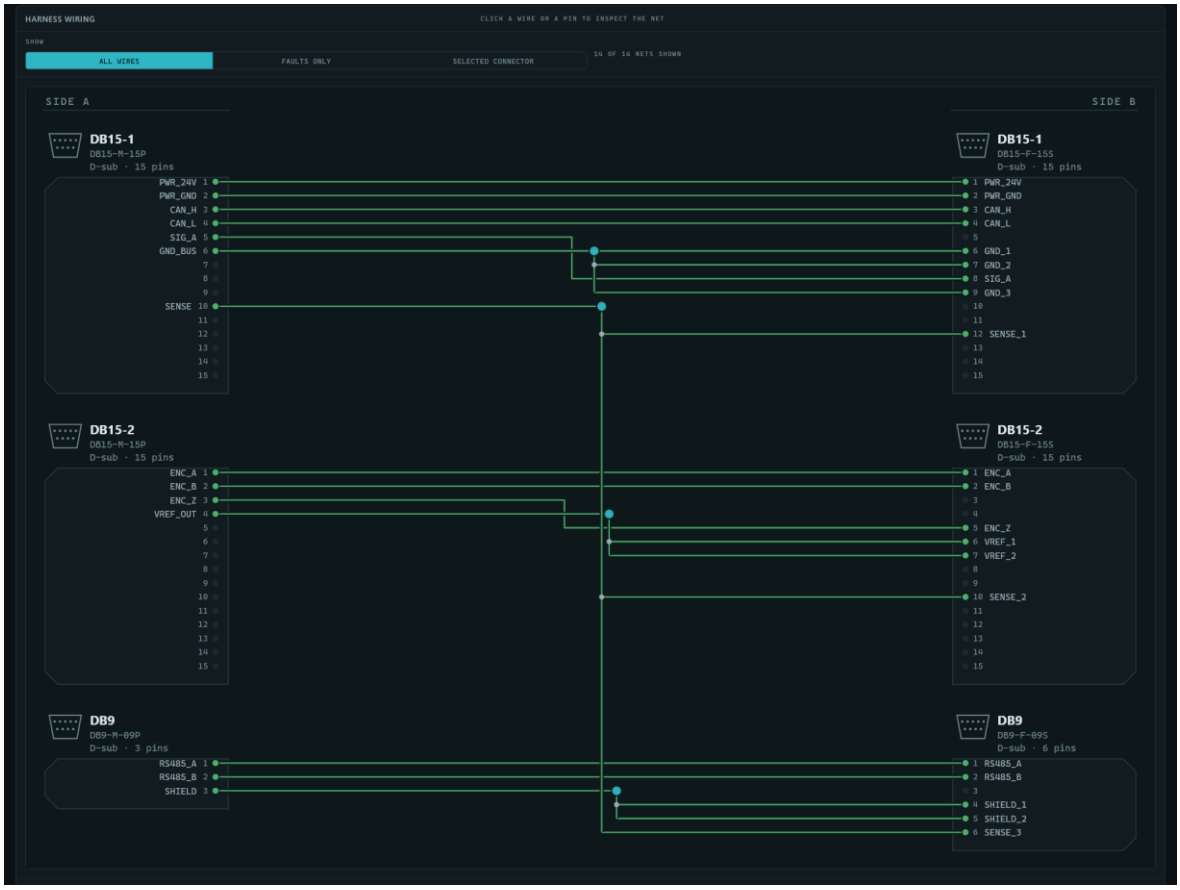


Figure 4: The harness wiring diagram: connectors, pins, orthogonally routed wires, Y joints and multi-drop nodes.

- **Show** filter: **All wires**, **Faults only** (nets currently in a failed state) or **Selected connector** (nets that touch the connector chosen in the Fixture connectors list; all nets if none is chosen). The count `n of m nets shown` is at the right.
- Each connector is drawn with a pictogram for its family (D-sub, circular, rectangular, terminal block, modular jack), its designator, mating part number, type and pin count, and numbered pin dots with the pin labels from the netlist.
- Wire and pin colours: green is passed, red is open or failed, grey is not tested. A **Y joint** (branch splice) and a **multi-drop node** are drawn as dots; an **I joint** (inline splice) is drawn as a mark on the wire.
- **Hover** over a wire or pin to highlight the net and see a tooltip with the net name, the pins at each end, the topology, the netlist's connector detail text and the state.
- **Click** a wire or pin to select the net: its pins are ringed and the Net inspector fills in. Click empty canvas to clear the selection.
- A legend under the diagram explains the colours and joint marks, and notes that Side A is the HI pins and Side B the LO pins.

Note: The firmware reports only pass or open for each pair, so the tooltip and the Net inspector show the state (`not tested`, `connected`, `OPEN (F06)`), not a voltage.

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7.7 Connection results table

Below the diagram (netlist mode only) the Connection results table lists every pair tested, one row per result as it arrives. Columns: Test #, Source, Part Number, Conn ID, Src Pin Label, Src Pin #, Dst Pin #, Dst Pin Label, Conn ID, Part Number, Destination and Status. Source and Destination are always blank. Status is **Connected** (green) or **Open** (red). The empty table reads `No results yet - run continuity to populate this table.`

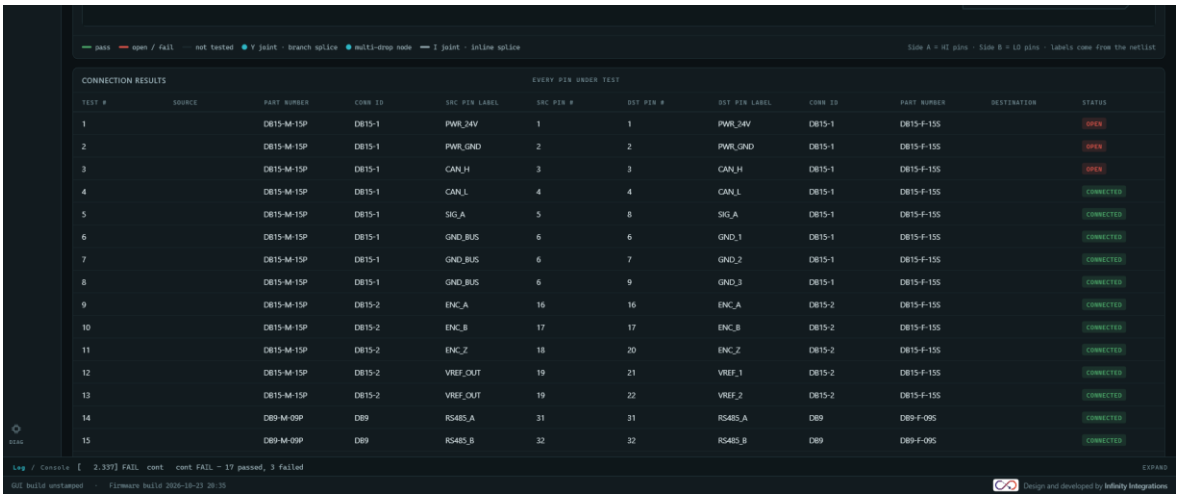


Figure 5: A continuity run with faults: open nets are drawn in red and listed in the results table.

7.8 Net inspector and Fixture connectors

Panel	Contents
Net inspector	Net name, From and To pins (connector, pin and label), Topology (`1:1 point to point`, `Y joint 1:N branch splice`, `Y joint N:1 merge`, `Multi-drop N to M wires on one node`, `I joint inline butt splice`), Detail (the netlist's connector detail text) and Result. A tag shows `nothing selected`, `open`, `connected` or `not tested`.
Fixture connectors	One row per connector: designator, label, a meta line (`Side A · D-sub · 15 way · board HS 001-015`) and a tag with its net count. Pressing a row toggles it as the selected connector, used by the Selected connector filter and dimming the other connectors in the diagram.
Discovered netlist (cross mode)	See Section 7.9.

Table 16: Continuity side panels

7.9 Cross continuity mode

Starting cross continuity clears the previous scan, sets the tiles to zero and disables saving until the scan completes. When a scan finds connections the Discovered netlist panel reads `N nets found across M pins. Save to load them as the working MTX netlist.` If it finds nothing it reads `No nets found. Check the harness is seated on J-MTX and run again.`

- **Save as MTX netlist** (enabled only after a scan that found pairs) sends the discovered pairs to the instrument and loads them as the MTX netlist.
- **Export CSV** (same condition) saves the pairs with the dialog title `Export discovered netlist`. The default file name is `discovered_netlist_<date>-<time>.csv`; the file has the header `HS pin,LS pin` and one line per pair.

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The Discovery panel of this mode (`Discovery - one to many`) lists nets built from the loaded net model rather than from the pairs the scan just found. For the reliable list of what a scan found, use the Auto Scan screen.

8 Resistance Screen

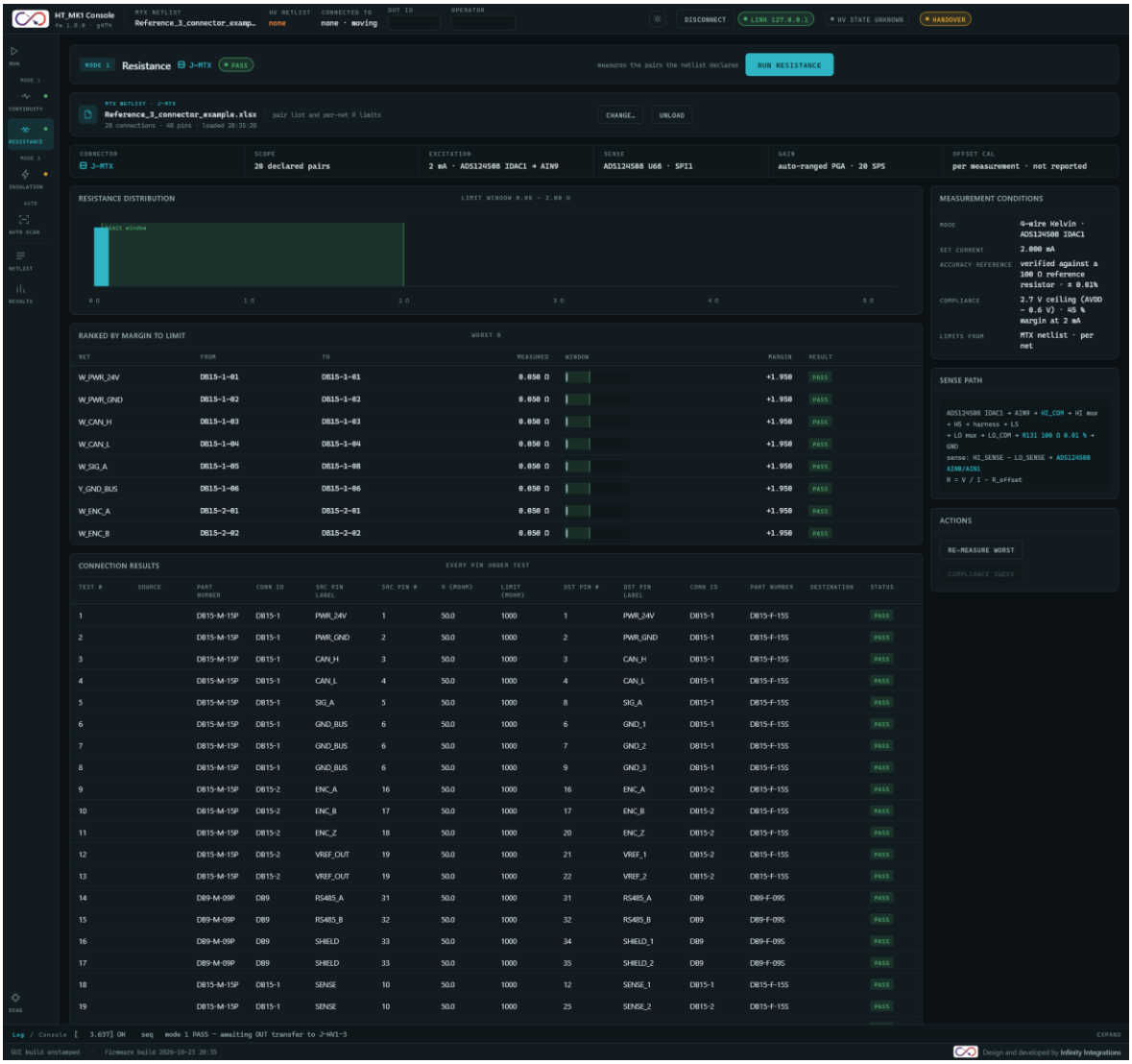


Figure 6: The Resistance screen after a passing run.

8.1 View bar, netlist strip and facts band

The view bar shows **Mode 1**, **Resistance**, **J-MTX**, the status pill, the reason text and **Run resistance**. The MTX netlist strip follows. The facts band shows **Connector** `J-MTX`, **Scope** (`N declared pairs` or `no netlist`) and **Offset cal** (`per measurement · not reported`).

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8.2 Main column

Panel	Contents
Resistance distribution	A histogram of the measured resistance of every net, with the caption `limit window 0.05 - 2.00 ohm`. This window is a GUI default per net, not a value read from the instrument.
Ranked by margin to limit	The worst 8 nets. Columns: Net, From, To, Measured (`x.xxx ohm`), Window (a mini bar showing the allowed window and a marker for the reading), Margin (limit minus reading, signed) and Result: **Pass** (green), **Marginal** (amber, margin under 0.2) or **F08 high** (red).
Connection results	Every pair tested. Columns: Test #, Source, Part Number, Conn ID, Src Pin Label, Src Pin #, R (mOhm), Limit (mOhm), Dst Pin #, Dst Pin Label, Conn ID, Part Number, Destination, Status (**Pass**, **F08 high**, **F08 low**). Resistance is shown to one decimal place. The limit is the instrument's value for that pair.

Table 17: Resistance main panels

8.3 Side column

Panel	Contents
Measurement conditions	**Mode** `4-wire Kelvin`; **Set current** `2.000 mA`; **Accuracy reference** (`verified against an ohm reference resistor` and the accuracy, or `not connected`); **Compliance** `2.7 V ceiling · 45 % margin at 2 mA`; **Limits from** `MTX netlist · per net` or `none loaded`.
Actions	**Re-measure worst** runs the full resistance test again (enabled when resistance is allowed). **Compliance sweep** is not available and stays disabled.

Table 18: Resistance side panels

9 Insulation Screen

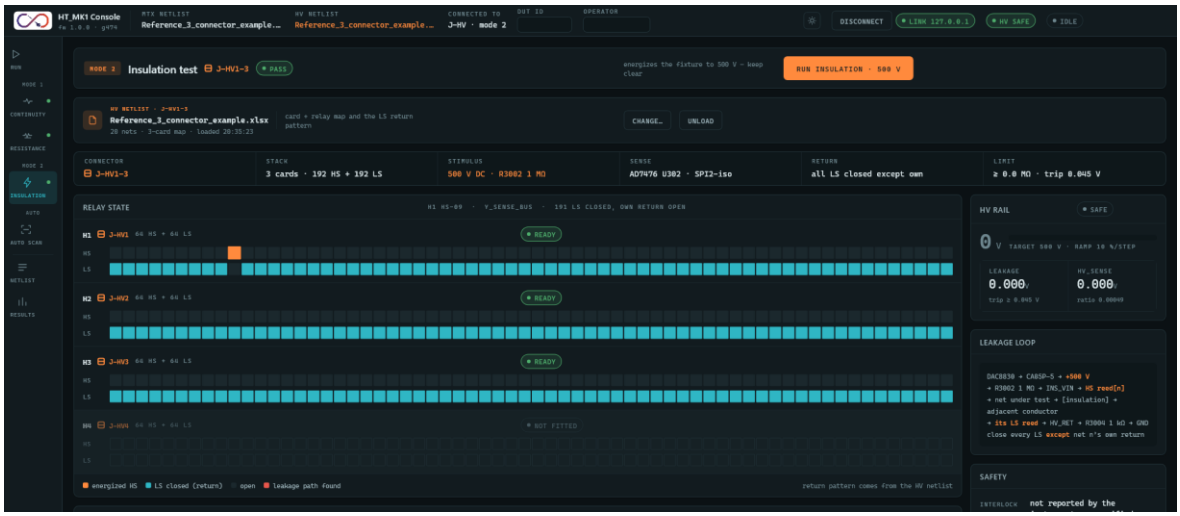


Figure 7: The Insulation screen after a run: relay state grid and HV rail.

9.1 View bar, netlist strip and facts band

The view bar shows **Mode 2**, **Insulation test**, the connector chip **J-HV1-N**, the status pill, the text `energizes the fixture to 500 V - keep clear` and the button **Run insulation · 500 V**. The button is enabled whenever no test is running. From J-MTX it opens the prerequisites dialog; from J-HV it goes to the HV netlist and verification dialogs (Section 13).

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The HV netlist strip shows the loaded file and an amber `stack mismatch` pill if its card count differs from the HV stack chosen on the Run screen. The facts band shows **Connector**, **Stack** (`N cards · 64N HS + 64N LS`), **Stimulus** `500 V DC`, **Return** `all LS closed except own` and **Limit** (the instrument's minimum insulation, `>= x.x MΩ`).

9.2 Relay state

Four rows, H1 to H4, one per possible HV card. Each has the card name, a connector chip (J-HV1 to J-HV4), `64 HS + 64 LS`, a pill (**Ready** green when fitted, **Not fitted** grey and dimmed otherwise) and two rows of 64 cells labelled HS and LS. The caption reads `idle - all relays open`, or while a net is under test the card, relay and net and the number of LS relays closed.

Cell colour	Meaning
HV colour (orange)	Energized HS relay (the net under test).
Blue	LS return relay closed.
Grey	Open.
Red	Leakage path found: the failing net's HS relay.
Outlined	The card is not fitted.

Table 19: Relay state colours

The grid follows the result of the most recently tested net. It is not a live readback of the relays.

9.3 Net results and Connection results

- **Net results** lists the worst 6 nets. Columns: Net, Conn, Card, HS relay, Insulation (`x.x MΩ`), Result (**Pass** or **Fail**).
- **Connection results** lists every net under test. Columns: Test #, Net, HV Card, HS Pin, Leak V, Insulation (MOhm), Limit (MOhm), Status. Leak V is derived from the rail voltage and the measured insulation, not sensed directly. The empty table reads `No results yet - run insulation to populate this table.`

9.4 HV rail and Safety panels

Panel	Contents
HV rail	The live rail voltage as a large number with a bar and the caption `target 500 V · ramp 10 %/step`, and a **Leakage** meter in volts that turns red at or above the fault threshold. The pill in the panel title shows the same HV state as the status bar HV pill (for example `HV Safe`).
Safety	Four facts: **Interlock** `not reported by the instrument · unverified`; **Switching** `cold · discharge before every relay change`; **Discharge** `passive · 10.05 MΩ · 200 ms wait`; **Return map** `HV netlist · per net`.

Table 20: Insulation side panels

- **Emergency discharge** is always enabled. It sends the abort command to the instrument, writes `ABORT sent` to the log and resets the GUI, exactly as the Abort & discharge button of the high-voltage banner does.
- **Relay self-test** is not available and stays disabled.

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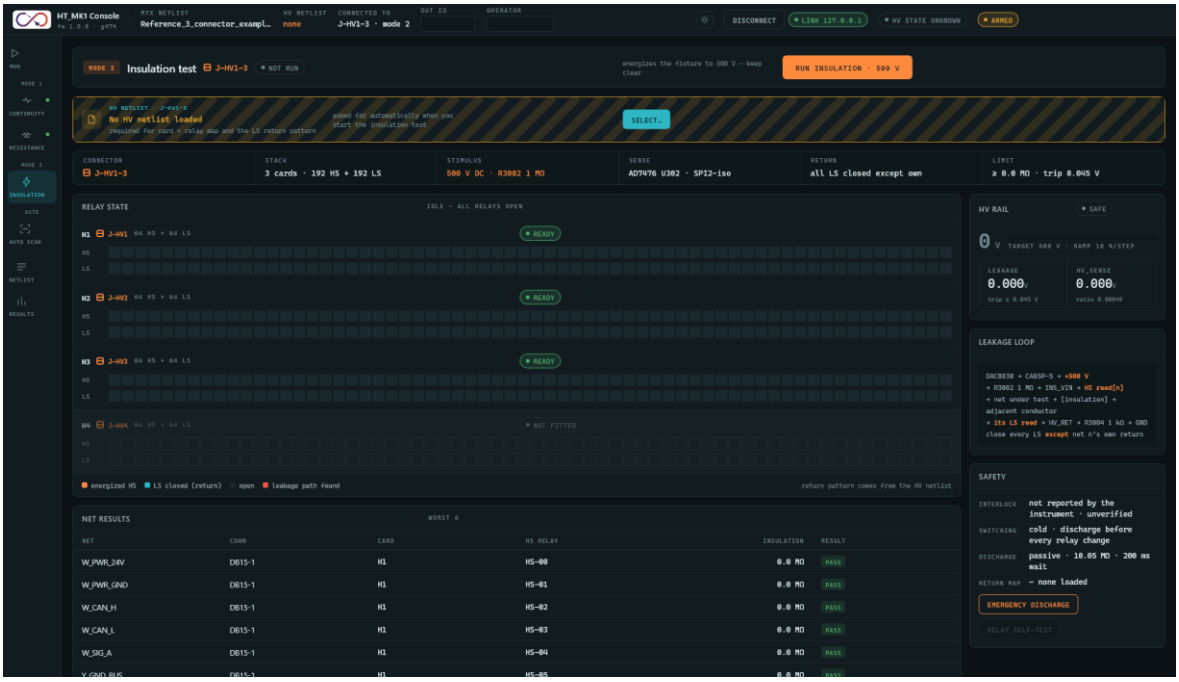


Figure 8: The Insulation screen before a run, showing the facts band, the relay grid and the Safety panel.

10 Auto Scan Screen

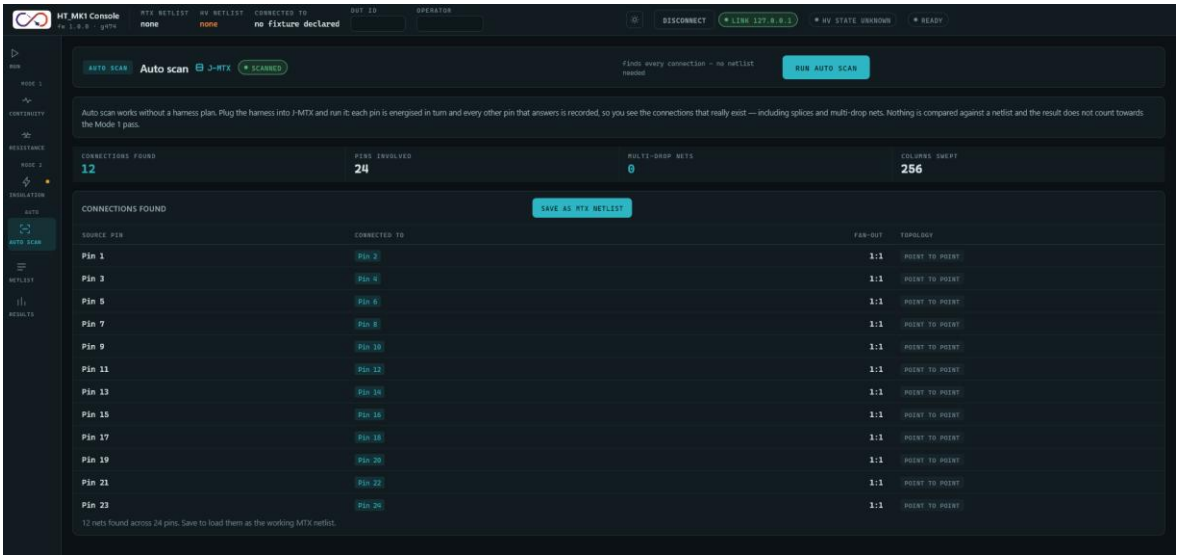


Figure 9: The Auto Scan screen after a scan.

Auto scan finds the connections that really exist on the harness without a netlist. Each pin is energized in turn and every other pin that answers is recorded, including splices and multi-drop nets. Nothing is compared with a netlist and the result does not count toward the Mode 1 pass.

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Item	Contents
View bar	Stage badge **Auto scan**, heading, **J-MTX**, a pill (`Not run`, `Scanning`, `Scanned`), a caption (`finds every connection - no netlist needed` or the reason it is blocked) and **Run auto scan** (disabled while a test runs or when the harness is on J-HV).
Description	The paragraph above.
Tiles	**Connections found**, **Pins involved**, **Multi-drop nets**, **Columns swept**. A dash until a scan has run.
Connections found	The table of results and the **Save as MTX netlist** button.

Table 21: Auto scan screen

- The table columns are **Source pin**, **Connected to**, **Fan-out** and **Topology**. There is one row per source pin (`Pin n`), a chip for each pin that answered, a fan-out such as `1:3`, and a tag **multi-drop** (more than one answer) or **point to point**. The first 300 source pins are shown, followed by `N more source pins not shown - save as a netlist to keep them all.`
- While scanning the table reads `Scanning - connections appear here as they are found.` Before any scan it reads `Run auto scan to find the connections on J-MTX.` After a scan: `N nets found across M pins. Save to load them as the working MTX netlist.`
- **Save as MTX netlist** (enabled only when pairs were found) sends the discovered pairs to the instrument and loads them as the MTX netlist.

When a scan finishes, the GUI discards the pass or fail it just recorded: the continuity result is cleared, the verdict returns to READY and the state pill to Idle. An auto scan therefore can never unlock insulation or raise the handover prompt. The continuity mode returns to Netlist when you leave the Auto Scan screen.

11 Netlist Screen

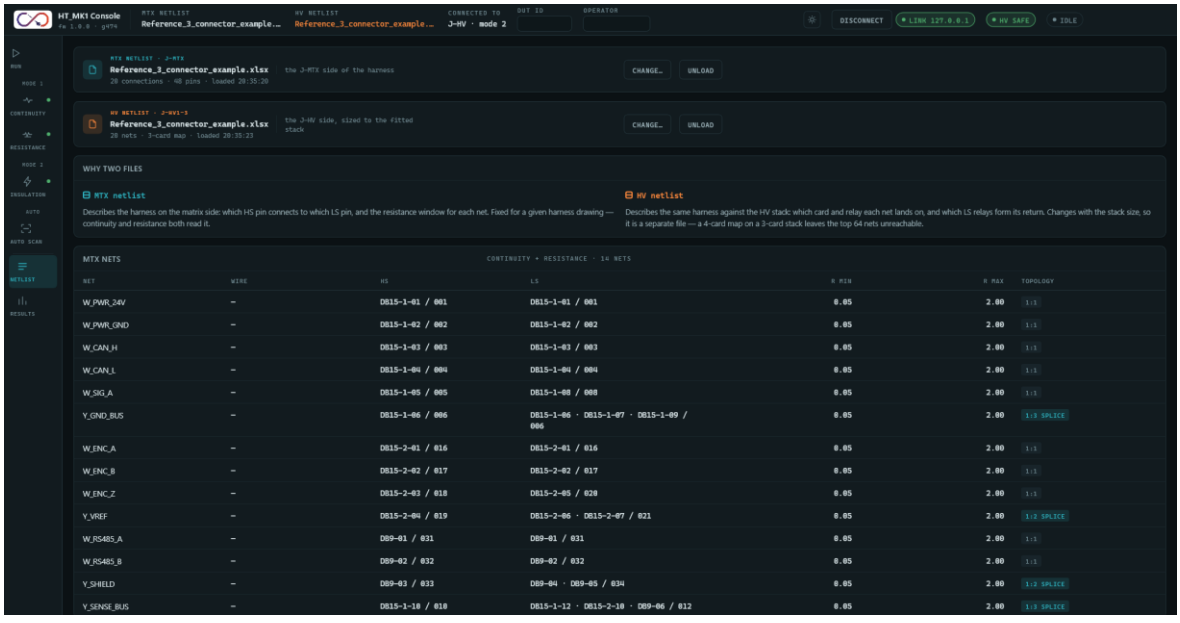


Figure 10: The Netlist screen: both netlists and the tables they define.

The Netlist screen shows the MTX netlist strip, the HV netlist strip, an explanation of why there are two files, and two tables. There is no save or export control on this screen.

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11.1 Netlist strip

The same strip appears on the Run, Continuity, Resistance, Insulation and Netlist screens. From left to right: a file icon, a kind line (`MTX netlist · J-MTX` or `HV netlist · J-HV1-N`), the file name, a meta line, an optional `stack mismatch` pill (HV strip only), a short caption saying what reads the file, and buttons.

Strip	State	Shown	Buttons
MTX	Not loaded	`No MTX netlist loaded; required for ...`; amber hatched border; caption `cross continuity can run without one and build it for you`	**Select...**
MTX	Loaded	File name; `N connections · M pins · loaded HH:MM:SS`, plus `built by cross scan` when it came from a scan	**Change...**, **Unload**
HV	Not loaded	`No HV netlist loaded`; caption `asked for automatically when you start the insulation test`	**Select...**
HV	Loaded	File name; `N nets · K-card map · loaded HH:MM:SS`	**Change...**, **Unload**

Table 22: Netlist strip states

- **Select...** and **Change...** open the Select the MTX netlist or Select the HV netlist dialog (Section 13).
- **Unload** on the MTX strip marks the netlist as unloaded in the GUI and blocks netlist-mode continuity and resistance (`MTX netlist unloaded - netlist continuity and resistance blocked`). It does not erase the instrument's own copy of the pairs.
- **Unload** on the HV strip blocks the insulation test until another HV netlist is selected.

11.2 MTX nets table

Title MTX nets, caption `continuity + resistance · N nets`. Columns: **Net**, **Wire** (always a dash), **HS** (connector, pin and the 3-digit HS pin), **LS**, **R min** and **R max** (GUI defaults 0.05 and 2.00) and **Topology** (`1:1`, `1:N splice`, `N:1 merge`, `multi-drop`, `inline splice`). The empty table reads `Load an MTX netlist to see the expected nets.`

11.3 HV nets table

Title HV nets, caption `N nets · M pins`. Columns: **Net**, **Card** (`H1`..), **HS relay** (`HS-nn`), **Return LS** (`all except LS-nn`), **Insul min** and **Reachable** (**yes** green, or **card not fitted** red when the net's card is beyond the chosen HV stack). The card and relay are calculated from the source pin: card = (pin - 1) / 64 and relay = (pin - 1) mod 64.

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12 Results Screen

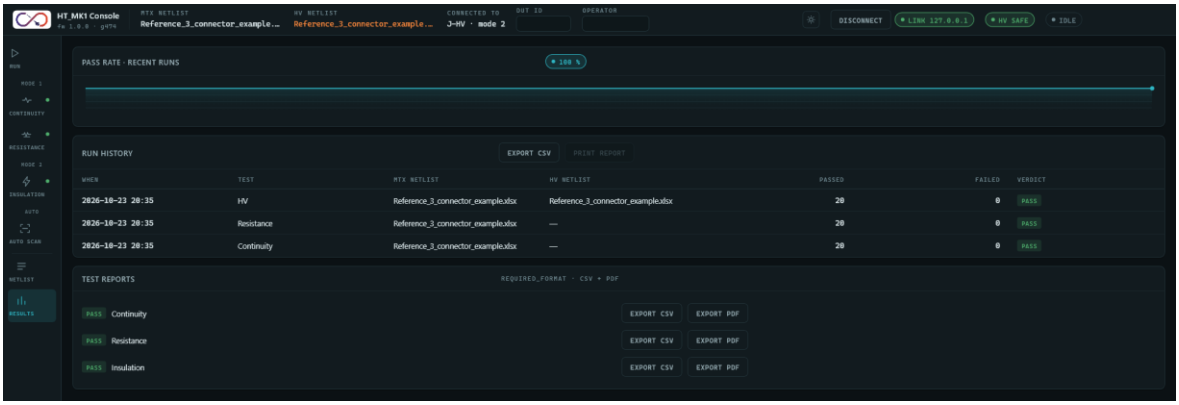


Figure 11: The Results screen: pass rate, run history and per-test reports.

12.1 Pass rate

The panel Pass rate · recent runs shows the overall pass percentage of the last 24 recorded runs in a pill, and a line of the rolling pass rate. With fewer than two runs it reads `Not enough runs recorded yet this session.`

12.2 Run history

One row is added for every completed run (continuity, resistance or insulation; an auto scan adds a Continuity row). The newest run is first. History is stored on the PC and is still there the next time the GUI starts (Section 17).

Column	Meaning
When	Date and time, `YYYY-MM-DD HH:MM`.
Test	Continuity, Resistance or HV.
MTX netlist, HV netlist	The file names in use, or a dash.
Passed, Failed	Counts for the run.
Verdict	**Pass** when Failed is zero, otherwise **Fail**.

Table 23: Run history columns

- **Export CSV** saves the whole history (title `Export run history`, default name `run_history_<date>-<time>.csv`). The columns are `Timestamp,Kind,MTX netlist,HV netlist,Passed,Failed,Verdict`, newest first, with Kind `cont`, `res` or `insul`. When there is nothing to export the log reads `no run history to export yet`.
- **Print report** is not available and stays disabled.

12.3 Test reports

The panel Test reports, with the caption `required_format · CSV + PDF`, has one row for each test that has completed with results in this session: Continuity (netlist mode only), Resistance and Insulation. Each row shows a **Pass** or **Fail** tag, the test name, **Export CSV** and **Export PDF**. A report is a snapshot of the most recent completed run of that kind, replaced when the next run of that kind completes. In addition, every completed run is saved automatically as a CSV and a PDF in the reports folder (Section 17), so a result is not

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lost if the GUI is closed before you press Export. With nothing to report the panel reads `No completed run this session to report on yet.`

The save dialog is titled `Export continuity report`, `Export resistance report` or `Export insulation report`. The suggested file name is `report_<date>-<time>_<continuity|resistance|insulation>.<csv|pdf>`.

12.4 CSV report content

Each CSV starts with comment lines (prefixed `#`), then a blank line, a column header and one row per result:

```
# Harness Tester Results [- Resistance | - HV Insulation]
# DUT ID,<DUT ID>
# Operator,<operator>
# Netlist,<MTX netlist name>
# (test-specific lines)
# Verdict,PASS|FAIL
# Date/Time,YYYY-MM-DD HH:MM:SS
```

Report	Extra header lines	Columns	Status values
Continuity	none	Test #, Source, Part Number, Conn ID, Src Pin Label, Src Pin #, Status, Dst Pin #, Dst Pin Label, Conn ID, Part Number, Destination	CONNECTED, NOT CONNECTED
Resistance	Excitation Current (mA); Resistance Limit (mOhm)	Test #, Source, Part Number, Conn ID, Src Pin Label, Src Pin #, Resistance (mOhm), Limit (mOhm), Status, Dst Pin #, Dst Pin Label, Conn ID, Part Number, Destination	PASS, FAIL_HIGH, FAIL_LOW
Insulation	HV Applied (V) = 500; Insulation Limit (MOhm)	Test #, Net, HV Card, HS Pin, Leak V, Insulation (MOhm), Limit (MOhm), Status	PASS, FAIL

Table 24: CSV report layouts

The verdict is PASS only when no result failed. A pin label defaults to `Pin n` when the netlist gives none. Leak V is an estimate derived from the rail voltage and the measured insulation.

12.5 PDF report content

A single flowing document: the company logo, the title (`Harness Tester Test Report`, `... -- Resistance` or `... -- HV Insulation`), the line `DUT: <id>` and the coloured line `VERDICT: PASS` or `FAIL`; metadata rows (Date/Time, Operator, Netlist, the test-specific lines above, Test Duration and the total pins or nets tested); a Summary table (totals and counts, tinted green and red); and the Results table with the same columns as the CSV, each row tinted green for CONNECTED or PASS and red otherwise.

13 Dialogs

Dialogs are drawn over the screen with a dimmed backdrop. Pressing Esc or clicking the backdrop dismisses them. The GUI has no error dialogs: every refusal, read error, timeout or link problem is reported only as a line in the log bar (Section 14). With the log bar collapsed, only the newest line is visible.

13.1 Select the HV netlist

Opens when insulation is started with no HV netlist loaded (the log says `HV start blocked - no HV netlist loaded, prompting operator`), and from Select... and Change... on any HV strip. The text explains that the HV netlist is separate because the relay map depends on the number of HV cards.

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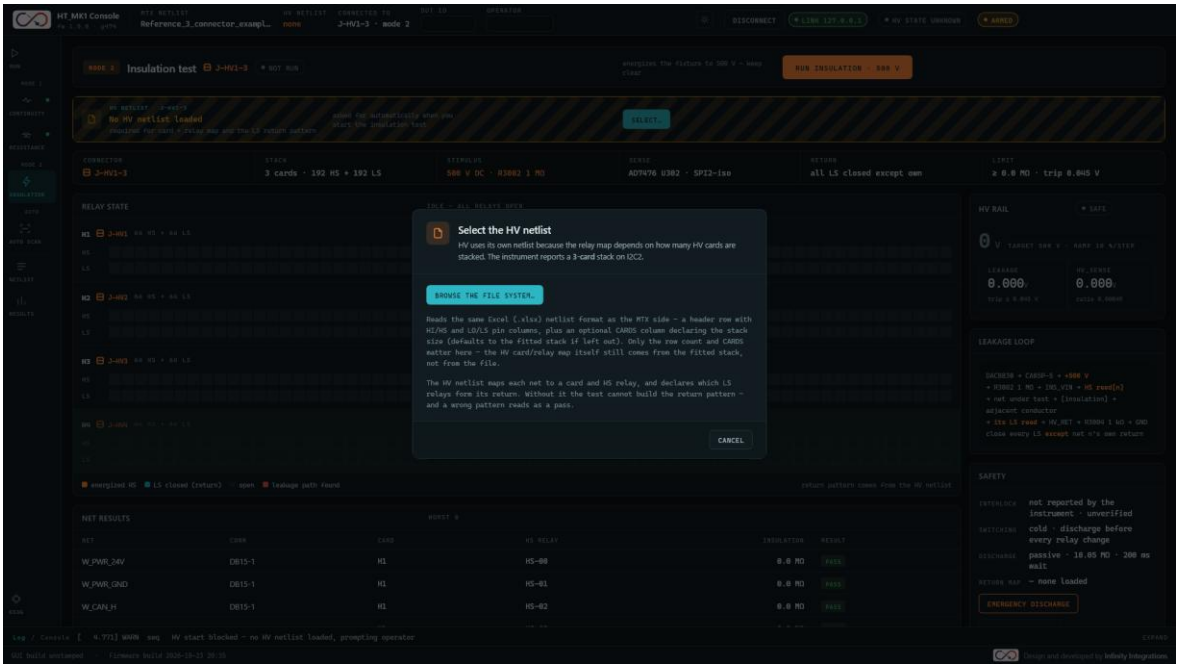


Figure 12: The Select the HV netlist dialog.

- **Browse the file system...** opens the Windows file dialog. The file is read on the PC and recorded in the GUI only; nothing is sent to the instrument. The GUI uses the number of rows as the net count and the largest value of the card-count column.
- On success the log reads `HV netlist loaded: <name> - N nets, K-card map.`; if the card count differs from the HV stack, `HV netlist declares K cards, S fitted - X nets unreachable`. The dialog closes and the verification dialog opens straight away.
- On a read error the log reads `could not read <name> - <message>` and the dialog stays open.
- **Cancel** closes the dialog and logs `HV start cancelled by operator`.

13.2 Select the MTX netlist

Opens from Select... and Change... on an MTX strip, and from the big Run Mode 1 button in netlist mode with no netlist loaded.

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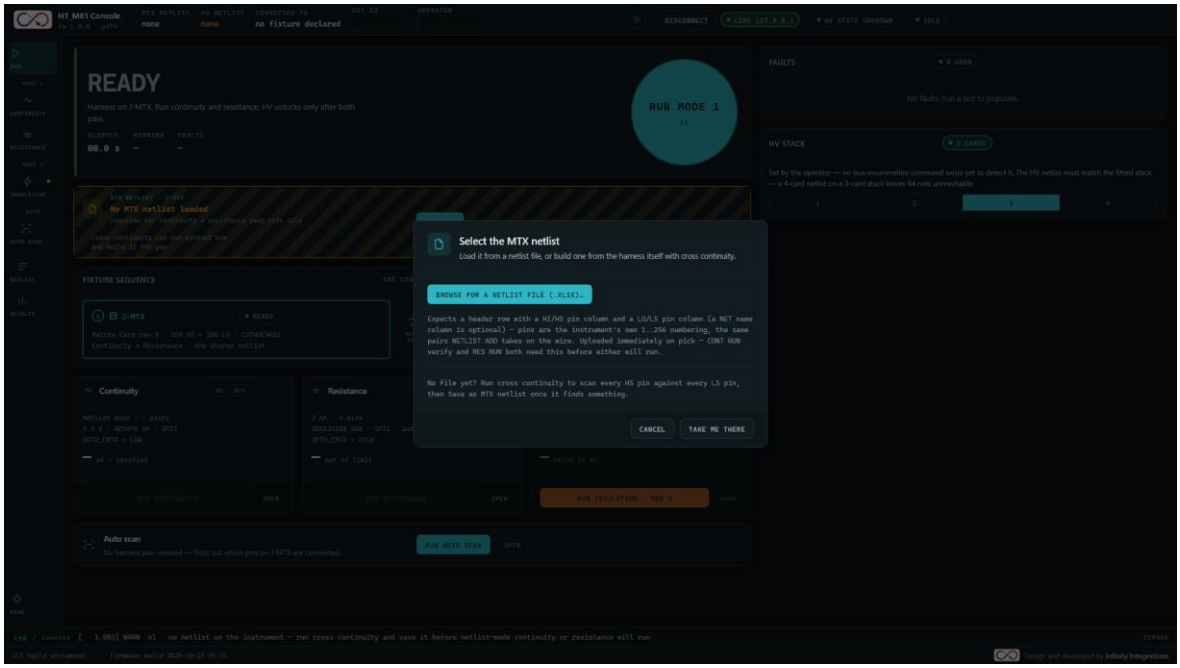


Figure 13: The Select the MTX netlist dialog.

- **Browse for a netlist file (.xlsx)...** opens the file dialog. The file is read, the connector layout is confirmed if the file has connector columns (Section 13.5), and the pairs are uploaded to the instrument.
- On success the log reads `MTX netlist loaded: <name> - N nets. Bound to continuity and resistance.` and `netlist uploaded to the instrument - N nets. Verify-mode continuity and resistance are now available.`
- If the instrument refuses the upload the log reads `netlist upload refused - <code> <text>` and the netlist is not loaded. A read error reads `could not read <name> - <message>`.
- **Take me there** closes the dialog, switches continuity to cross mode and opens the Continuity screen, for when there is no file yet.
- **Cancel** closes the dialog.

13.3 Confirm before energizing to 500 V

Opens on every insulation start, after the HV netlist is available. The tick box is unticked each time.

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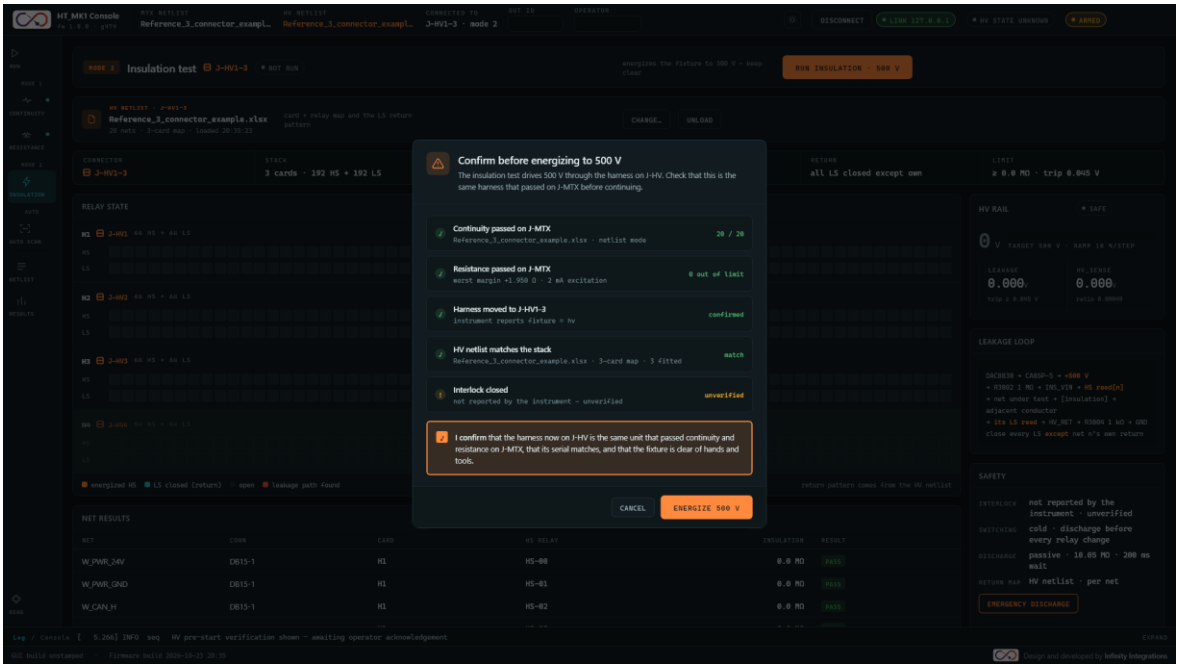


Figure 14: The pre-start verification dialog with the confirmation box ticked and Energize 500 V enabled.

Check row	Detail	Value
Continuity passed on J-MTX	MTX file name and mode	`N / N` (green), `skipped` (amber) or `not passed` (red)
Resistance passed on J-MTX	Worst margin and `2 mA excitation`	`0 out of limit`, `skipped` or `not passed`
Harness moved to J-HV1-N	`instrument reports fixture = hv` or `instrument has not confirmed the harness moved`	`confirmed` (green) or `not confirmed` (amber)
HV netlist matches the stack	File name, card map, cards fitted	`match` (green) or `N nets unreachable` (amber)
Interlock closed	`not reported by the instrument - unverified`	`unverified` (amber, always)

Table 25: Pre-start verification rows

- The confirmation text reads: `I confirm that the harness now on J-HV is the same unit that passed continuity and resistance on J-MTX, that its serial matches, and that the fixture is clear of hands and tools.` Clicking anywhere on the box toggles the tick.
- **Energize 500 V** is disabled until the box is ticked. Pressing it closes the dialog, logs `operator acknowledged HV pre-start checks - energizing`, then arms the instrument and starts the insulation run. If the arm command is refused, the log reads `arm refused - <code> <text>` and nothing starts.
- The check rows are informational. A red or amber row does not block the button; only the tick box does. Read every row before ticking.
- **Cancel** closes the dialog and logs `HV start cancelled by operator`.

13.4 Run continuity and resistance first? / Move the harness to J-HV

Opens when any Run insulation control is pressed while the harness is not on J-HV. The log reads `insulation requested - asking the operator about the Mode 1 pre-tests`.

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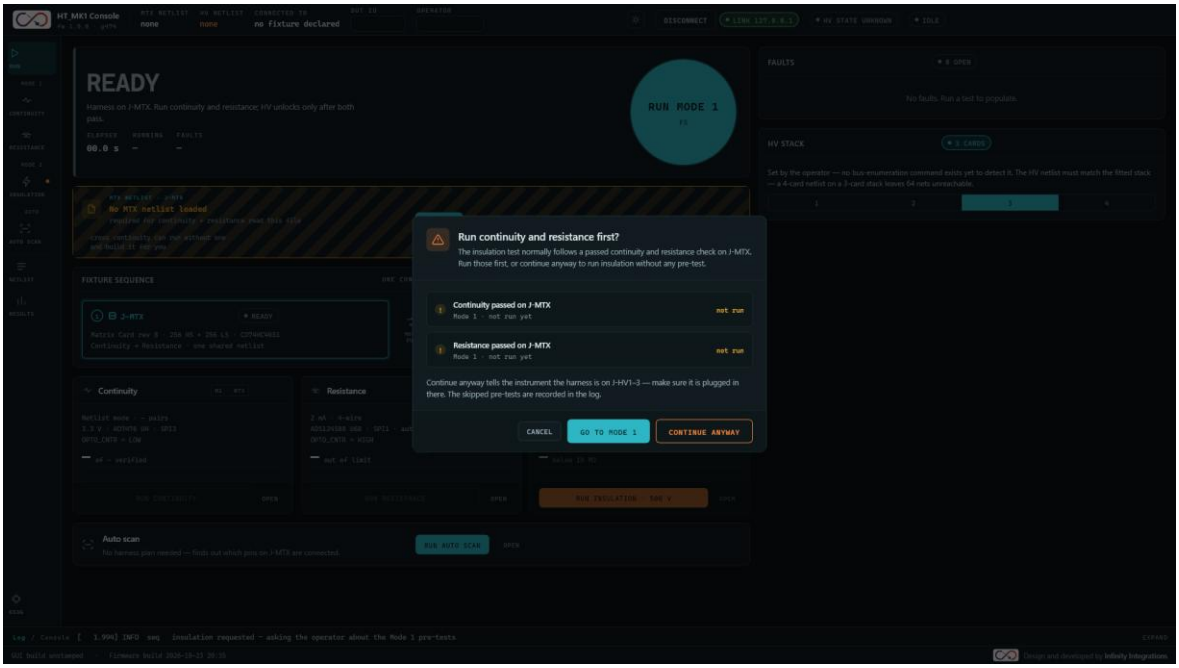


Figure 15: The prerequisites dialog when Mode 1 has not been completed.

Mode 1 state	Title	Buttons
Continuity and resistance both passed	`Move the harness to J-HV1-N`	**Cancel**, **Harness moved - continue**
Anything else	`Run continuity and resistance first?`	**Cancel**, **Go to Mode 1**, **Continue anyway**

Table 26: Prerequisites dialog

- Two check rows show `Continuity passed on J-MTX` and `Resistance passed on J-MTX` with the detail `Mode 1 · completed on J-MTX`, `failed on J-MTX` or `not run yet` and the value `passed`, `failed` or `not run`.
- **Go to Mode 1** closes the prompt and opens the Continuity screen.
- **Harness moved - continue** and **Continue anyway** both tell the instrument the harness is on J-HV (a refusal is logged as `fixture hv refused - <code> <text>`), then go on to the HV netlist dialog if needed and the verification dialog.
- **Continue anyway** runs insulation without a completed Mode 1. The log records `insulation started WITHOUT a completed continuity + resistance pass - operator chose Continue anyway`, and the verification dialog shows `skipped` instead of `not passed`.

13.5 Confirm the connector layout

Opens after an MTX netlist that has Conn ID or Part Number columns is read. The guessed layout is already applied behind the dialog, and **Cancel** reverts to the layout that was active before.

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Figure 16: The Confirm the connector layout dialog.

- One row per connector: its designator, `Side A` or `Side B` and its pin count; a five-way selector for the shape (**D-sub**, **Circular**, **Rect**, **Terminal**, **Modular**); and a text box with the part number or friendly name. A blank name restores the designator. Pin counts, identifiers and order are not editable.
- The footnote explains that a connector with fewer pins than expected usually means its trailing pins were never wired in the file. That is normal, not an error.
- **Apply** keeps the layout with your shapes and names. **Cancel** (or Esc) reverts it.
- If a connector has more pins wired than its part number allows, the log warns `Side A connector "<id>" has N pins wired but its part number (<pn>) has M. ....`

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13.6 Windows file dialogs

Opening a netlist uses the dialog titled `Select a netlist file`. Its filter lists `.xlsx` and `.xls`, but only `.xlsx` files can be read. Saving uses the titles `Export run history`, `Export discovered netlist`, `Export continuity report`, `Export resistance report` and `Export insulation report`.

14 Log Bar and Footer

14.1 Log bar

The log bar runs along the bottom of every screen. Its head (always visible) shows the tab **Log** and the newest log line in the form `[time] LEVEL source message`. The **Expand** toggle at the right opens a 150 pixel scrolling list of all lines since start-up, scrolled to the newest; **Collapse** closes it.

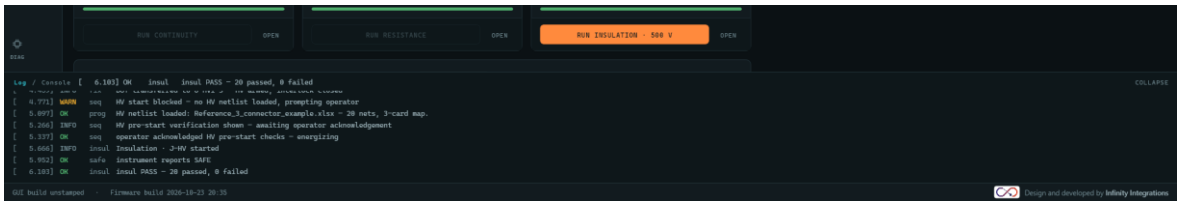


Figure 17: The log bar expanded.

Level	Colour	Used for
INFO	Grey	Ordinary events (connected, netlist read, fixture change).
OK	Green	Completed actions and passing results.
WARN	Amber	Things to check: missing netlist, mismatches, skipped pre-tests.
FAIL	Red	Refusals, read errors, link problems and failing results.
HV	HV colour	High-voltage events.

Table 27: Log levels

The time stamp in each line is a running counter of the GUI, not a wall-clock time. The list is held in memory only and has no filter, search or clear control; the permanent record is the session log (Section 17). The first line after start-up reads `HT_MK1 console started - not yet connected`.

14.2 Footer

The footer shows `GUI build <date> <time>` and, once the instrument has answered, `Firmware build <date> <time>`. At the right are the logo and the credit line `Design and developed by` followed by the company name.

15 Keyboard Shortcuts

Key	Action
F5	Same as pressing the big round button, only when no test is running. It is not blocked by the button being disabled, so pressing F5 during the handover resets the GUI.
Esc	If the connector layout dialog is open: cancels it and reverts the layout. Otherwise, if any other dialog is open: closes it. Otherwise, if a test is running: sends the abort command to the instrument and resets the GUI. Otherwise: resets the GUI.

Table 28: Keyboard shortcuts

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There are no other shortcuts.

16 Operating Sequences and States

16.1 Normal sequence

1. **Connect.** Choose the port and press Connect. The state is Idle and Connected to reads `J-MTX · mode 1`.
2. **Load an MTX netlist** (or build one with Cross continuity or Auto scan). Netlist-mode continuity and resistance are blocked without one.
3. **Mode 1.** Press Run Mode 1 or Run continuity. Results arrive pair by pair, and progress drives the card bar and the Elapsed value. Then press Run resistance. Each test passes only when its failed count is zero.
4. **Handover.** When both pass the verdict reads MOVE DUT and the handover panel appears. If either fails, the verdict reads FAIL, the harness stays on J-MTX and HV is not attempted.
5. **Move the harness** to J-HV and press Harness moved - unlock HV. The state becomes Armed.
6. **Start insulation.** Press Run Insulation. Select an HV netlist if asked, then read the verification dialog, tick the confirmation and press Energize 500 V. The instrument ramps the rail; at 50 V or more the high-voltage banner replaces the status bar.
7. **Results stream in** net by net. When the run ends the instrument discharges the rail and reports safe. Wait for the green HV Safe pill before touching the harness.

16.2 When a run finishes

The GUI builds the report snapshot, saves it (Section 17), adds a run-history line, colours the test card and navigation dot, clears the running state and writes a log line such as `cont PASS - 20 passed, 0 failed`. When insulation ends the verdict headline changes to PASS or FAIL with the line `Insulation passed on J-HV1-N. Wait for HV Safe before handling the harness.` (or `Insulation failed ...`), and the stage 2 pill and the state pill show Pass or Fail. Pressing the big button then resets to READY.

16.3 Abort, Stop and Reset

Control	What it does
Abort & discharge (high-voltage banner)	Sends ABORT to the instrument, logs `ABORT sent`, then resets the GUI.
Emergency discharge (Insulation screen)	The same as Abort & discharge. It is always enabled.
Stop button and Esc (a test is running)	Send ABORT to the instrument, log `ABORT sent`, then reset the GUI. The instrument stops after the measurement in progress, removes high voltage and reports safe.
Reset button and Esc (no test running)	Reset the GUI only. They clear results, tallies, faults and the relay display, mark the harness as back on J-MTX, close dialogs and show READY.

Table 29: Stopping a test

The instrument answers an abort with the end-of-run report of the part that ran. The GUI discards that partial result (`<test> run aborted: partial result discarded`): an aborted run gets no verdict, no history row and no report, so a half-finished run can never be recorded as a pass.

Warning: Wait for the green HV Safe indicator after any abort before touching the harness.

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A reset does not clear the loaded netlists, the connector layout, the HV stack size, the report snapshots or the stored run history.

16.4 Safe to handle

The GUI shows the green **HV Safe** pill only when the instrument has sent its own safe report and nothing since has contradicted it. A run ending, silence from the instrument or the order of events never makes the GUI show safe. After connecting, the state is unknown until the instrument reports safe. A reported rail above zero, a running test, the armed state or a fault all return the GUI to live or unknown. When the safe report arrives the log reads `instrument reports SAFE`.

16.5 Faults

A latched fault appears as the red state pill Fault, a log line with the fault code and text, and the **Clear Fault** button at the right of the status bar. Every run request is refused while a fault is latched; if a run ends while in fault the log reads `<test> refused - fault latched; clear the fault and retry`. Remove the cause, then press **Clear Fault**: the instrument forces everything safe, clears the latch (a refusal is logged as `fault clear refused - <code> <text>`) and the state returns to Idle. A fault is never cleared automatically.

16.6 Loss of the link

The link is declared lost when the instrument does not reply to a command within 2 seconds, when no bytes arrive for 5 seconds, or when the port is closed or errors (for example the cable is pulled or the board resets). The GUI then:

- shows `No link`, `Link lost` and `State unknown` and Connected to `link lost`;
- removes the high-voltage banner and clears the running test, keeping the results already on screen;
- writes `link lost - <detail> (state unknown)` to the log, for example `command timeout after 2.0s`, `no traffic for 5.0s` or `port closed by instrument`;
- does not reconnect by itself. Press Connect again.

Link loss never shows as safe. While the link is down every command fails with `link lost - state unknown; reconnect required`. Treat the system as unsafe to handle until the link is back and HV Safe is shown.

17 Files and Folders

The GUI writes to `%LOCALAPPDATA%\HT_MK1`. If that location is not writable it uses another per-user location.

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What	Location	Format
Session log	`sessions\session-<date>-<time>.log` (a new file at each Connect)	Text, one line per command, reply and event, with time stamps, flushed as written.
Run history	`history\run_history.jsonl`	JSON lines, one per completed run, written immediately and read again at every start.
Reports	`reports\report_<date>-<time>_<test>.csv` and `.pdf` (one pair per completed run)	The CSV and PDF test reports, saved automatically.
Theme	`theme`	The word `dark` or `light`.
Exports	Wherever you save them	CSV or PDF.

Table 30: Files written by the GUI

Not saved between sessions: the log bar, DUT ID and Operator, the netlist file names and details (the instrument keeps only the pin pairs), the HV stack size and the connector layout.

Appendix A: Netlist File Format

The same reader is used for both the MTX and the HV netlist. The accepted file type is Microsoft Excel `.xlsx` only. CSV files are not accepted as netlists. CSV is used only for exports.

- The **first sheet** is read; its name does not matter. Row 1 is the header row. Completely blank rows are skipped.
- Header names are matched without regard to case. Columns the GUI does not recognise are ignored.
- HI and LO must be whole numbers from 1 to 256. They are two independent address spaces (the instrument's HS and LS multiplexer banks). HI equal to LO is allowed. A repeated pair is an error.
- Rows with the same Net name form one net: one source with several rows is a Y branch; several sources into one destination is a merge; several at both ends is a multi-drop node; one row is a plain wire. Rows with no Net name stand alone and are named `NET_001`, `NET_002` and so on.

Field	Required	Accepted header texts
HI pin (Side A, HS)	Yes	hi, hi pin, hs, hs pin, high, from, from pin, pin a, pin1, pin 1, src pin #, src pin, source pin
LO pin (Side B, LS)	Yes	lo, lo pin, ls, ls pin, low, to, to pin, pin b, pin2, pin 2, dst pin #, dst pin, destination pin
Net name	No	net, net name, name, signal
HV card count	No (HV netlist)	card, cards, hv card, stack
Side A connector	No	conn id, connector, conn
Side A part number	No	part number, part #, part no
Side B connector	No	conn id b, connector b, conn b
Side B part number	No	part number b, part # b, part no b
Source pin label	No	src pin label, source pin label, src label, pin label
Destination pin label	No	dst pin label, destination pin label, dst label
Connector detail	No	connector detail, wire detail, detail, joint

Table 31: Netlist columns

Connector detail text

The text is read by keyword, without regard to case. A detail containing `splice`, `inline` or `butt` declares an inline (I) splice. A detail containing `y branch`, `y-branch` or `y joint` declares a Y joint. Anything else

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(`straight`, `crossed pin 5 -> 8`, blank) declares no joint. Structure outranks text: a net with several rows is a Y or multi-drop whatever the detail says. Only the inline splice, which cannot be seen from structure, is taken from the text. The detail is shown in the Net inspector and the diagram tooltip and is never sent to the instrument.

How the connector layout is guessed

When Conn ID columns exist, Conn ID and Part Number describe the Side A connectors (HI pins) and Conn ID B and Part Number B the Side B connectors (LO pins). Each end is analysed on its own, so a designator used at both ends becomes two connectors. Connectors are ordered by their lowest pin; each connector's block runs from just after the previous block to just before the next connector's lowest pin, and the last connector runs to the highest pin used. Each side numbers its pins from 1. The family is guessed from the part number:

Family	Part number contains
Terminal block	TERMINAL, SCREW, PHOENIX, WAGO, MSTB, DINKLE, TBLOCK, TB-, TB_
Modular jack	RJ45, RJ-45, RJ11, RJ-11, RJ12, MODULAR, ETHERNET, USB, 8P8C
D-sub	DSUB, D-SUB, DB, or DA, DB, DC, DD, DE, HD followed by digits
Rectangular	RECT, CPC, MOLEX, DEUTSCH, JST (and anything not matched)
Circular	CIRC, MS3, MS27, 38999, MIL-DTL, M8, M12, M16, M23, M27, AMPHENOL

Table 32: Connector family keywords

The pin count stated by the part number is used only as a check. Without Conn ID columns the GUI uses a built-in layout, and pins outside it get connectors marked as unassigned. Confirm or correct the result in the Confirm the connector layout dialog.

What is sent to the instrument

Only the pin pairs of the MTX netlist are sent (`NETLIST BEGIN`, one `NETLIST ADD` per pair, `NETLIST END`). The instrument holds them in RAM and they are gone after every power-up; on connect the GUI reads them back, but names, labels and details are not held by the instrument and are lost until the file is loaded again. The HV netlist is not sent at all: the card and relay for each net are calculated from the source pin.

Appendix B: Start-up Options

The programs can be started from a command line with these options. For normal use no option is needed: start the program, choose the port and press Connect.

Option	Meaning
--serial <port>	Use that COM port and connect at start-up, for example `--serial COM7`. `--serial auto` picks the ST-LINK Virtual COM Port when exactly one is attached.
--baud <n>	Serial rate. Default 115200, 8 data bits, no parity, 1 stop bit.
--list-ports	Print the serial ports the PC can see and exit.
--log-dir <path>	Folder for the session logs.
--sim	Run the built-in simulator instead of an instrument (the Demo program). Cannot be combined with `--serial`.
--scenario <name>	With --sim: `pass`, `opens_shorts`, `res_fail`, `insul_fail` or `disconnect`.
--nets <n>	With --sim: size of the simulated harness (default 12).
--host <addr>, --port <n>	Connect over TCP instead of a serial port.
--help	Print the options and exit.

Table 33: Start-up options

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With no options the program starts without connecting and asks you to choose a port in the status bar. The Demo program also shows a Demo harness panel on the Run screen (Good mode 1 or Faulty mode 1) that changes what the simulator reports.

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